

Exam Simulation - May 31, 2021

Exam Duration: 10 minutes

Questions:

- 4 Questions from PT course X 22.5 puncte
- 1 Required question (10 points when the answer is YES)
- 1 Required question regarding entering your name, group and TA

Question types:

- Radio-button: one correct answer;
 - Checkbox: one or more (including all) correct answers; no partial credit;
- All code examples in Java

Exam mark: Number of points /10

...

Points: 55/100

1. Do you want the 10 default points? *
(10/10 Points)

☒ YES ✓

2. Consider the following code fragment:

```
public class Operation {  
    ...  
    public int sum( int a, int b ) { return a + b; }  
    ...  
}
```

```
...  
Operation operation = new Operation();  
Class aClass = operation.getClass();
```

Which of the following answers is correct for the dynamic invocation of method sum using the reflection technique

Answer a:

```
Method sumMethod = aClass.getMethod("sum", new Class[] {});  
sumMethod.invoke(4, 5);
```

Answer b:

```
Method sumMethod = aClass.getMethod("sum", new Class[] {int.class, int.class});  
sumMethod.invoke(operation, new Object[] {4, 5});
```

Answer c:

```
Method sumMethod = aClass.getMethod("sum", new Class[] {int.class, int.class});  
sumMethod.invoke(4, 5);  
(22.5/22.5 Points)
```

☐ Answer a

☐ Answer c

☒ Answer b ✓



3. Consider a class A and a class B. Class A has an instance variable of type B, a constructor and methods of type get and set.
In which of the following situations, a dependency may be injected in class A?
(0/22.5 Points)

- ☐ Another class C creates an instance of B and then instantiates an object of type A passing the object instance of B as parameter. ✓
- ☐ Class A creates an object of class B
- ☐ Another class C creates an instance of class B and then invokes the set method of class A using an existing object of class A. ✓
- ☐ Class B creates an instance of class A.
- ☐ Another class C creates an instance of class B and then invokes the get method of class A using an existing object of class A.



4. Consider the interface defined below:

```
public interface AnInterface {
    public static final boolean FLAG = true;
    default void defaultOperation();

    private void privateOperation() {
        System.out.println("I am a private operation");
    }
}
```

Which of the following statements is correct?

(0/22.5 Points)

- ☐ The given code does not include any mistake
- ☐ The interface AnInterface is incorrect defined because the default method must be defined (implemented) ✓
- ☐ The interface AnInterface is incorrect defined because an interface cannot define constant values
- ☐ The interface AnInterface is incorrect defined because it is not allowed to declare private methods in interfaces

5. Enter the following: <Group>, <Name>, <Surname>, <Year when you have taken the course>, <TA name> *

Enter your answer

6. Which of the following statements about conversion of reference types are true?
(22.5/22.5 Points)

- ☐ Downcasting is allowed at compile time with an explicit cast but may generate exceptions in execution ✓
- ☐ Downcasting doesn't need explicit conversions
- ☐ Downcasting represents the conversion of a supertype to one of its subtype ✓
- ☐ Widening represents the conversion of a supertype to one of its subtype

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